

CURRICULUM VITAE

Toghrul Karimov, PhD

Postdoctoral Researcher

Max Planck Institute for Software Systems, Saarbrücken, Germany

Contact and identifiers

Full name (as in publications)	Toghrul Karimov
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Scientific field	Computer science

Profile summary

Toghrul Karimov is a postdoctoral researcher at the Max Planck Institute for Software Systems in Saarbrücken, where he has worked since February 2024. He was a PhD student there from 2019 to 2024, supervised by Joël Ouaknine. The thesis, Algorithmic verification of linear dynamical systems, was submitted at Saarland University in 2024 and received the grade summa cum laude. He holds an MCompSci in computer science from the University of Oxford (2015–2019), with first-class honors. His papers concern decidability and verification of linear dynamical systems.

Education

Years	Degree	Field / thesis title	Institution, city, country
2019–2024	PhD	Algorithmic verification of linear dynamical systems. Supervisor: Joël Ouaknine. Grade: summa cum laude. Saarland University, 2024	Max Planck Institute for Software Systems, Saarbrücken, Germany; degree at Saarland University
2015–2019	MCompSci	Computer science. First-class honors	University of Oxford, Oxford, United Kingdom

Academic and professional positions

Years	Position	Department / unit	Institution, city, country
February 2024–present	Postdoctoral Researcher		Max Planck Institute for Software Systems, Saarbrücken, Germany

Publications**Book chapters**

1. Aghamov, R.; Baier, C.; **Karimov, T.**; Nieuwveld, J.; Ouaknine, J.; Vahanwala, M. Model checking Markov chains as distribution transformers. In: Principles of Verification: Cycling the Probabilistic Landscape. Lecture Notes in Computer Science, 15261, 2024.
2. **Karimov, T.**; Kelmendi, E.; Ouaknine, J.; Worrell, J. What’s decidable about discrete linear dynamical systems? In: Principles of System Design: Thomas A. Henzinger Festschrift. Lecture Notes in Computer Science, 13660, 2022.

Peer-reviewed journal articles

1. Berthé, V.; **Karimov**, T.; Nieuwveld, J.; Ouaknine, J.; Vahanwala, M.; Worrell, J. The monadic theory of toric words. *Theoretical Computer Science*, 1025.

Conference papers and abstracts

1. **Karimov**, T.; Luca, F.; Nieuwveld, J.; Ouaknine, J.; Worrell, J. On the decidability of Presburger arithmetic expanded with powers. *SODA 2025*, to appear.
2. Berthé, V.; **Karimov**, T.; Nieuwveld, J.; Ouaknine, J.; Vahanwala, M.; Worrell, J. On the decidability of monadic second-order logic with arithmetic predicates. *Proceedings of LICS 2024*. Distinguished Paper Award.
3. Aghamov, R.; Baier, C.; **Karimov**, T.; Piribauer, J.; Ouaknine, J. Linear dynamical systems with continuous weight functions. *Proceedings of HSCC 2024*. ACM SIGBED Best Paper Award.
4. **Karimov**, T.; Kelmendi, E.; Nieuwveld, J.; Ouaknine, J.; Worrell, J. The power of positivity. *Proceedings of LICS 2023*.
5. Baier, C.; Funke, F.; Jantsch, S.; **Karimov**, T.; Lefauchaux, E.; Ouaknine, J.; Purser, D.; Whiteland, M.; Worrell, J. Parameter synthesis for parametric probabilistic dynamical systems and prefix-independent specifications. *Proceedings of CONCUR 2022*.
6. D’Costa, J.; **Karimov**, T.; Majumdar, R.; Ouaknine, J.; Salamati, M.; Worrell, J. The pseudo-reachability problem for diagonalisable linear dynamical systems. *Proceedings of MFCS 2022*.
7. **Karimov**, T.; Lefauchaux, E.; Ouaknine, J.; Purser, D.; Worrell, J.; Whiteland, M. What’s decidable about linear loops? *Proceedings of the ACM on Programming Languages*, 6, POPL 2022.
8. Baier, C.; Funke, F.; Jantsch, S.; **Karimov**, T.; Lefauchaux, E.; Luca, F.; Ouaknine, J.; Purser, D.; Whiteland, M.; Worrell, J. The orbit problem for parametric linear dynamical systems. *Proceedings of CONCUR 2021*.
9. D’Costa, J.; **Karimov**, T.; Majumdar, R.; Ouaknine, J.; Salamati, M.; Soudjani, S.; Worrell, J. The pseudo-Skolem problem is decidable. *Proceedings of MFCS 2021*.
10. Almagor, S.; **Karimov**, T.; Kelmendi, E.; Ouaknine, J.; Worrell, J. Deciding ω -regular properties on linear recurrence sequences. *Proceedings of the ACM on Programming Languages*, 5, POPL 2021.
11. Baier, C.; Funke, F.; Jantsch, S.; **Karimov**, T.; Lefauchaux, E.; Ouaknine, J.; Pouly, A.; Purser, D.; Whiteland, M. Reachability in dynamical systems with rounding. *Proceedings of FSTTCS 2020*.
12. **Karimov**, T.; Ouaknine, J.; Worrell, J. On LTL model-checking for low-dimensional discrete linear dynamical systems. *Proceedings of MFCS 2020*.

Grants and research projects

Years	Project title	Funding body	Your role
	CPEC mini-project, travel to Oxford to work with Amaury Pouly on reachability for linear time-invariant systems	DFG grant 389792660, TRR 248	

Awards, honors and fellowships

Year	Award / honor	Awarding body
2024	Distinguished Paper Award, LICS	LICS
2024	ACM SIGBED Best Paper Award, HSCC	HSCC
2016–2019	Keble College Scholarship	Keble College, University of Oxford
2015–2019	Scholarship covering the full cost of study at the University of Oxford	Ministry of Education of Azerbaijan

Teaching

Years	Course	Level	Institution
Winter 2022	Topics in algorithmic dynamical systems theory	Teaching assistant	Saarland University
Summer 2020	Automata and sequences	Teaching assistant	Saarland University

Invited talks and conference participation

Year	Event, place	Type	Title of talk
2024	Symbolic dynamics and arithmetic expansions workshop, Roscoff, France; logic seminar, Stellenbosch University, South Africa	Talk	Ode to o-minimality
2023	LICS 2023	Talk	The power of positivity
2023	Bellairs workshop, Barbados	Talk	The model-checking problem for linear dynamical systems
2022	MFCS 2022 and RP 2022	Talk	The pseudo-reachability problem for diagonalisable affine dynamical systems
2021	MFCS 2021	Talk	The pseudo-Skolem problem is decidable
2021	Monsoon Math 2021	Talk	Invariants and impossibility: from geometric constructions to solving polynomial equations
2021	POPL 2021	Talk	Deciding ω -regular properties on linear recurrence sequences
2020	MPI-SWS	Lightning talk	On verification of linear dynamical systems
2020	MFCS 2020	Talk	On LTL model-checking for low-dimensional discrete linear dynamical systems